

Lab 8: Code Review

Motivations

- Code review is standard in industry--it happens before any commit to the main branch
- Reading, commenting and critiquing someone's approach, broaden everyone perspective
 - we learn each time we read carefully code
 - we see different qualities in other's work
 - we reflect on our own work
- Deep work/thinking on paper for everyone benefit

Code Review

At the function level the code should be [clean](#), as synthesize below.

I hope that is Martin's site; I encourage you to read through it in your own time. [Uncle Bob](#) is an important figure of the software engineering industry, pioneer of the agile method among other things.

Clean code

From the article link above, I selected the important content for the micro-component of our activity.

Essence of Clean Code

Clean code is like **a well-organized kitchen**. You know where everything is, ingredients are labeled, and tools are easy to reach. Anyone can walk in and cook a meal without calling the chef.

At the function level, clean code is:

- Easy to understand
- Easy to maintain and extend
- Resistant to bugs
- (Easy to test) -- dev mode for use

Core Principles of Clean Code

- Meaningful Names — Like labelling a jar *sugar* instead of *white powder*
- Small Functions — Like having a *coffee maker* instead of a multi-appliance mess
- Minimal Comments — Like a map that's so clear, it doesn't need instructions
- No Duplication — Like reusing a recipe card instead of rewriting the same steps
- Consistent Formatting — Like aligned books on a shelf — clean and organized

Review Process Description

At the meta-level, you should follow important/applicable steps of this [guide](#)

Steps: 1., 2., 3.1-3 3.6, 4., 5. and 8.

When reviewing your partner's code, try asking:

- If I was new here, would I understand this?
- Are game's structures we studied identifiable?
- How much messy/spaghetti code is out there? [messy code](#)
- Will this design still make sense in 6 months?
- If we doubled the load, would this hold up?

Code Review: Steps and Fill-in below

Name: _____ Partner: _____

- Skim through the printed code (10 mins)
- Read in details most of the functions and **annotate/write specific comments** on the actual printed code, that will help you for the life-review. (30 mins)
- Write in the three boxes below, in bullets form for each of the topic (15 mins)

Preparing for code review

1. The good: list principles that are well followed; code parts that are crystal clear

2. The not as good: Need some work , i.e. what is missing and/or suggestions to improve

3. Difference: which sections/mechanisms you would like to learn from? or is too confusing? and will need to spend some time on

Code review

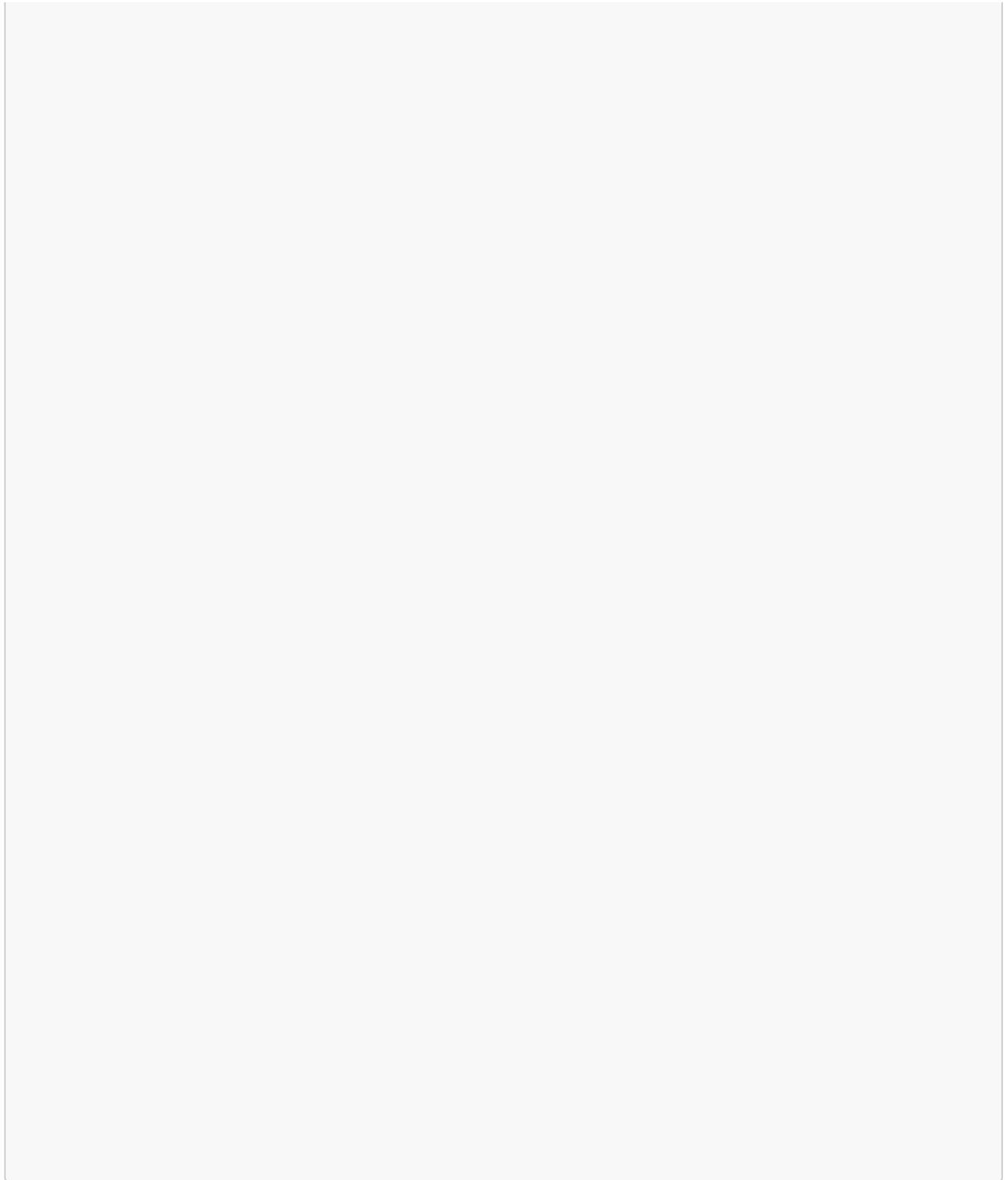
When both of you are ready

- each in turn do the code review of the other (15 mins each)

After code review

- Describe four reflections on this activity
 - two on how it changes your thoughts about coding in general
 - two on how you view your own project in particular

Reflections



Submission

- Hand in
 - this sheet and
 - annotated code before the end of this lab